

Wine Quality Classification Project Report

Using PCA, LDA, and SVM for Red vs. White Wine Classification

Dataset: wine_quality_merged.csv

Report purpose: explain the project idea, preprocessing steps, visualizations, numerical results, and model performance.

1. Project Idea and Objective

This project works on the Wine Quality dataset. The main goal is to classify the wine type as either red wine or white wine using the chemical features available in the dataset. The original type column was encoded into numerical labels where red = 0 and white = 1. After that, several methods were trained and compared: PCA for dimensionality reduction and analysis, LDA implemented from scratch and using sklearn, and SVM with an RBF kernel implemented from scratch and using sklearn.

A strong point in this project is that it does not only depend on ready-made library functions. It also compares from-scratch implementations with sklearn implementations, which shows an understanding of the algorithm steps rather than only running built-in models.

2. Dataset Overview

Item	Value / Information
Rows before cleaning	6,497 rows
Columns before cleaning	13 columns
Numerical features used	12 features
Target variable	type: red or white
Missing values	0 in all columns
Duplicate rows	1,177 duplicate rows
Rows after removing duplicates	5,320 rows
Train/test split	80% training and 20% testing
Training size	4,256 samples
Testing size	1,064 samples

The dataset did not contain missing values, so imputation was not needed. However, it contained 1,177 duplicate rows. These duplicates were removed to reduce bias and avoid training the model on repeated copies of the same samples.

3. Short Statistical Description

Feature	Mean	Std	Min	Max
fixed acidity	7.215	1.296	3.800	15.900
volatile acidity	0.340	0.165	0.080	1.580
residual sugar	5.443	4.758	0.600	65.800
chlorides	0.056	0.035	0.009	0.611
free sulfur dioxide	30.525	17.749	1.000	289.000
total sulfur dioxide	115.745	56.522	6.000	440.000
alcohol	10.492	1.193	8.000	14.900
quality	5.818	0.873	3.000	9.000

The statistics show that some features have very large ranges, such as total sulfur dioxide and free sulfur dioxide, while other columns have smaller ranges, such as pH and density. Therefore, using StandardScaler before PCA and SVM was an appropriate decision because scale-sensitive algorithms can be affected when numerically larger features dominate the calculations.

4. Exploratory Data Analysis (EDA)

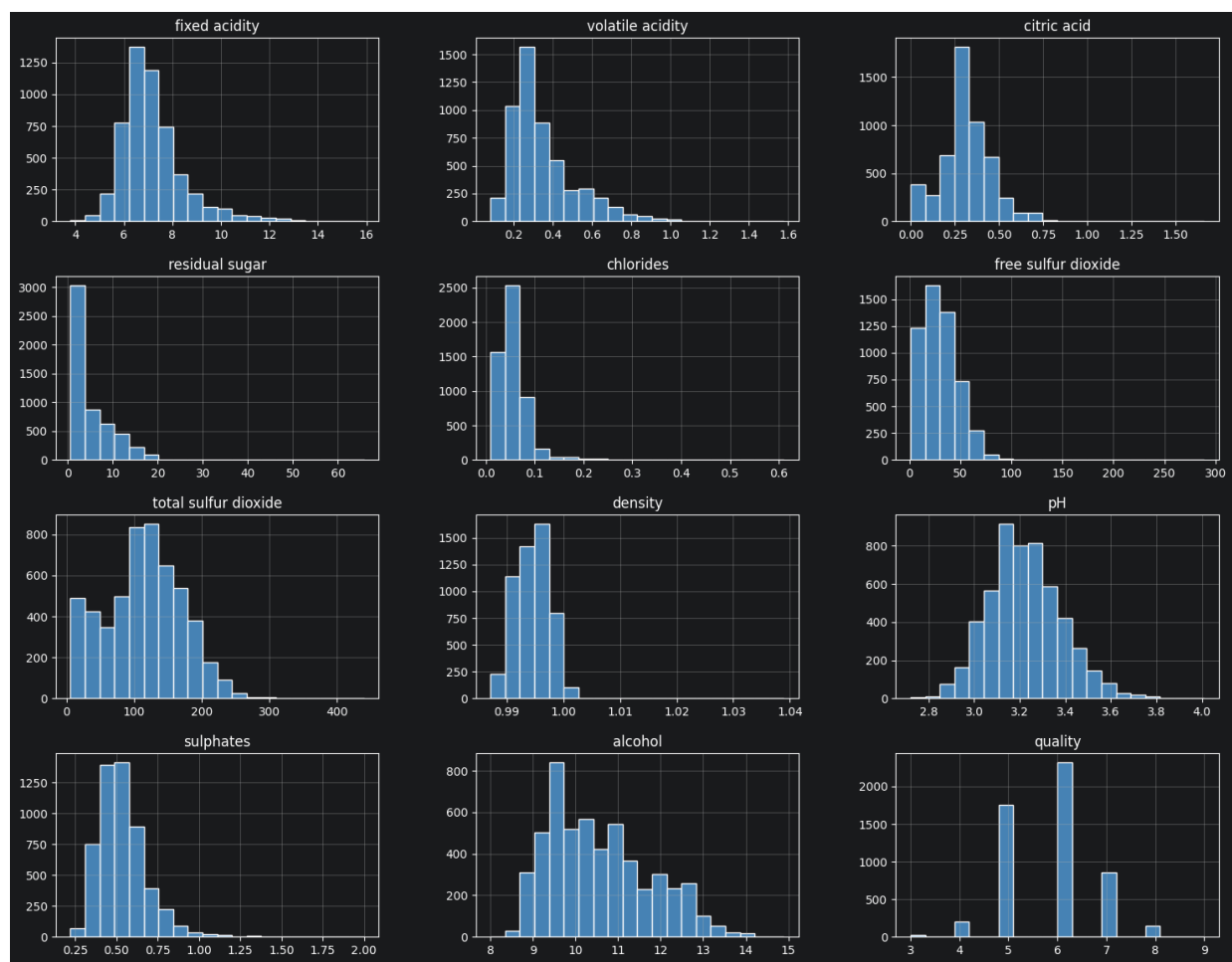


Figure 1: Histograms of the numerical features. These plots show the distribution of each feature and help identify skewness and possible outliers.

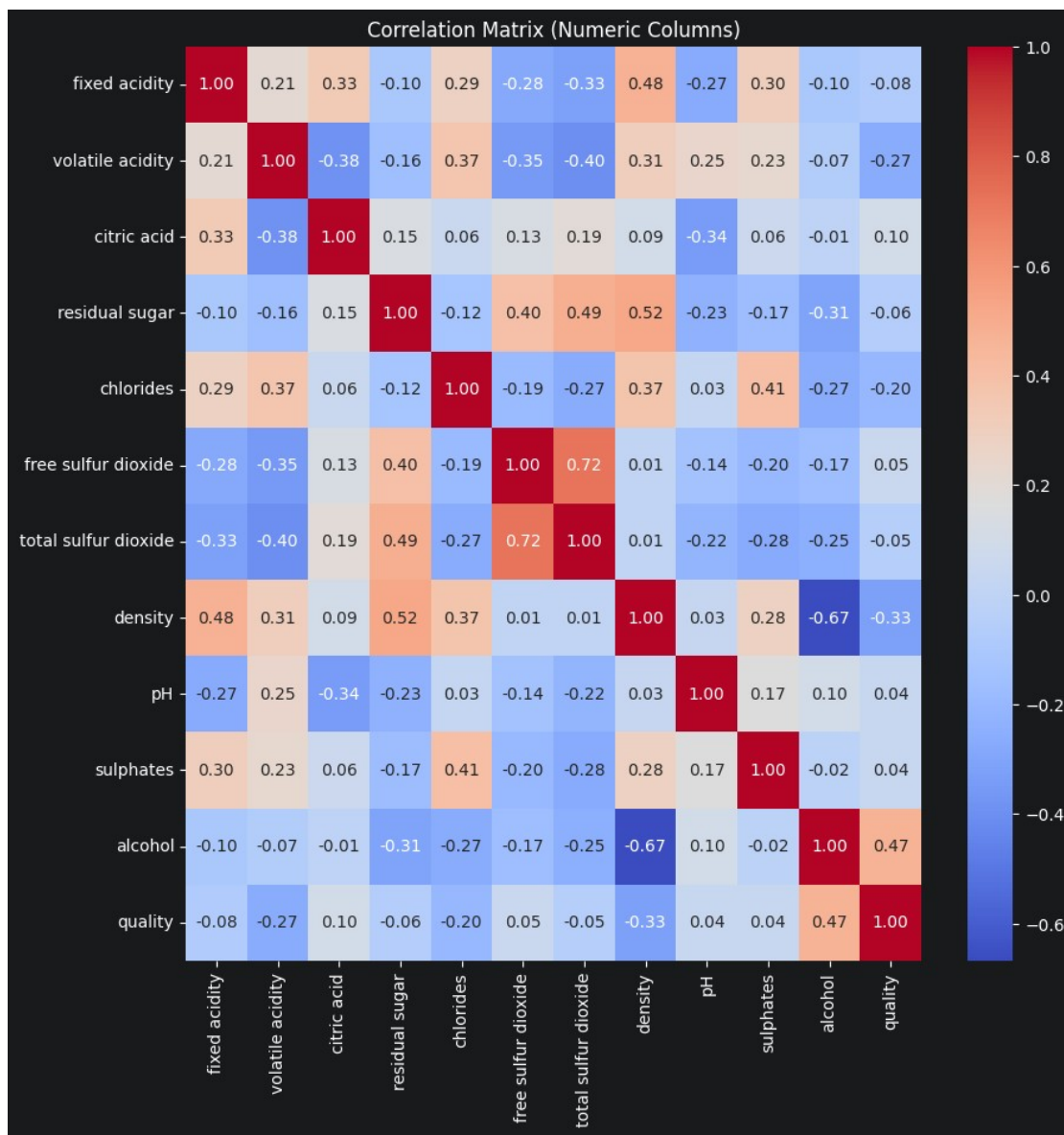


Figure 2: Correlation matrix. The heatmap shows relationships between numerical features and helps identify strongly related variables.

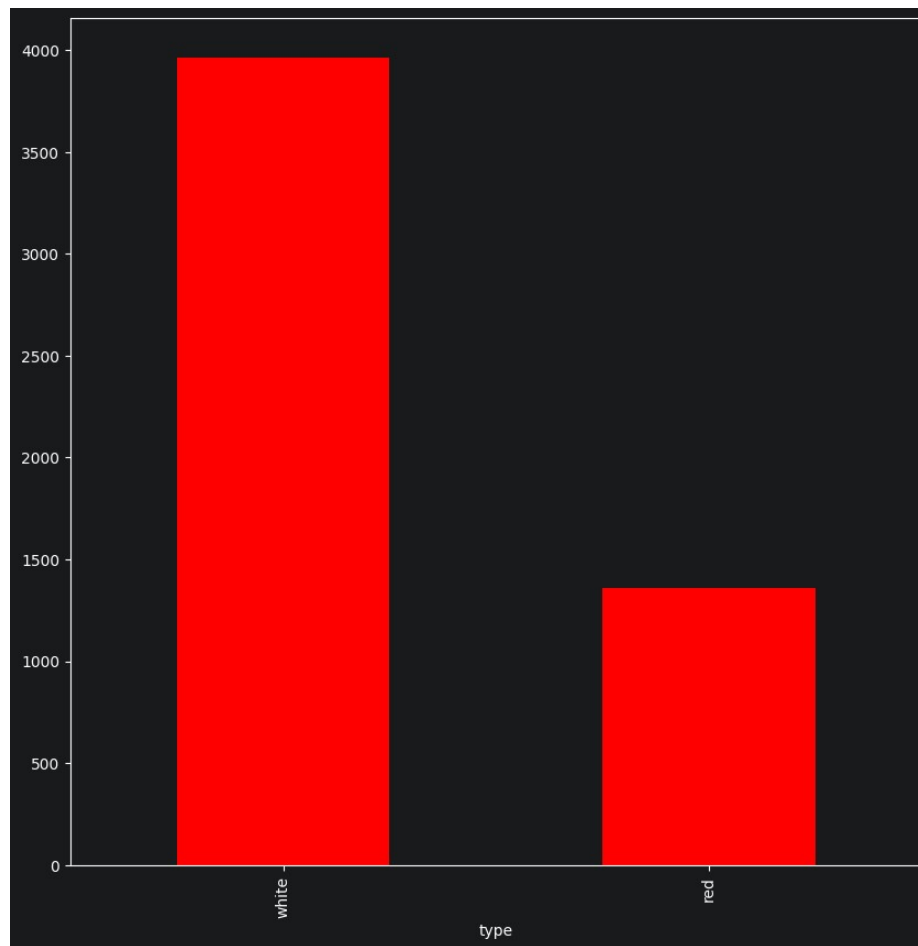


Figure 3: Distribution of the type column. The dataset contains two classes, red and white, with more white wine samples than red wine samples.

The histograms are useful because several features are not perfectly normally distributed and some contain extreme values. The correlation heatmap is also important because it shows relationships between chemical properties, which may help the classification models separate red wine from white wine.

5. Preprocessing and Data Preparation

The data was prepared through several steps: removing duplicate rows, separating the features X from the target y, converting the type column into type_encoded, splitting the data with stratification to preserve the red/white class ratio in both training and testing sets, and applying StandardScaler for models that are sensitive to feature scale. Clipping was also applied using the training-set mean and standard deviation at the mean $\pm 3 \times \text{std}$ boundaries to reduce the effect of extreme values before LDA.

Step	Reason
drop_duplicates	Reduces repeated samples and prevents model bias
type encoding	Converts red/white labels into 0/1 values for modeling
stratified train_test_split	Preserves class distribution in training and testing sets
StandardScaler	Unifies feature scale, especially for PCA and SVM
Outlier clipping	Reduces the effect of extreme values on LDA

6. PCA From Scratch and Comparison With sklearn

PCA was implemented from scratch by calculating the mean, centering the data, computing the covariance matrix, extracting eigenvalues and eigenvectors, sorting them in descending order, and then projecting the data onto the most important principal components. The result was compared with sklearn PCA, and the explained variance ratios were almost identical, which supports that the from-scratch implementation is correct.

Principal Component	Explained Variance Ratio
PC1	0.2523
PC2	0.2180
PC3	0.1408
PC4	0.0895
PC5	0.0712
PC6	0.0539
PC7	0.0458
PC8	0.0423
PC9	0.0396
PC10	0.0248

The first 10 principal components preserve about 97.8% of the total variance. The first component explains 25.23%, and the second component explains 21.80%, so the first two components together explain about 47.03%. According to the scree plot, around 9 components are enough to reach approximately 95% cumulative explained variance.

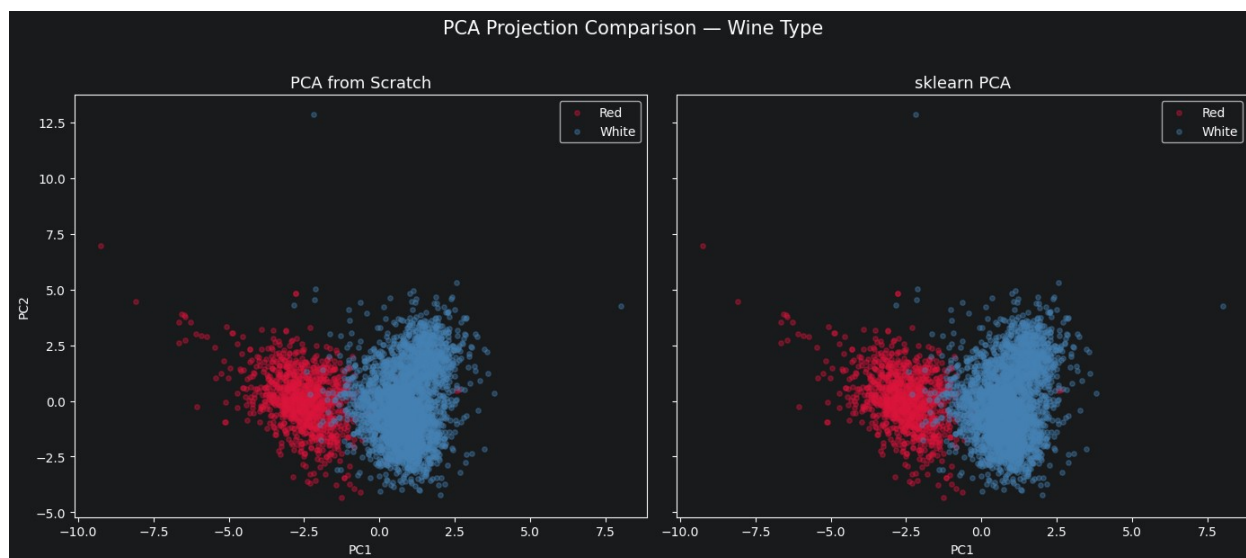


Figure 4: Comparison between PCA projection from scratch and sklearn PCA. The similarity between the plots shows that the manual PCA calculations match sklearn closely.

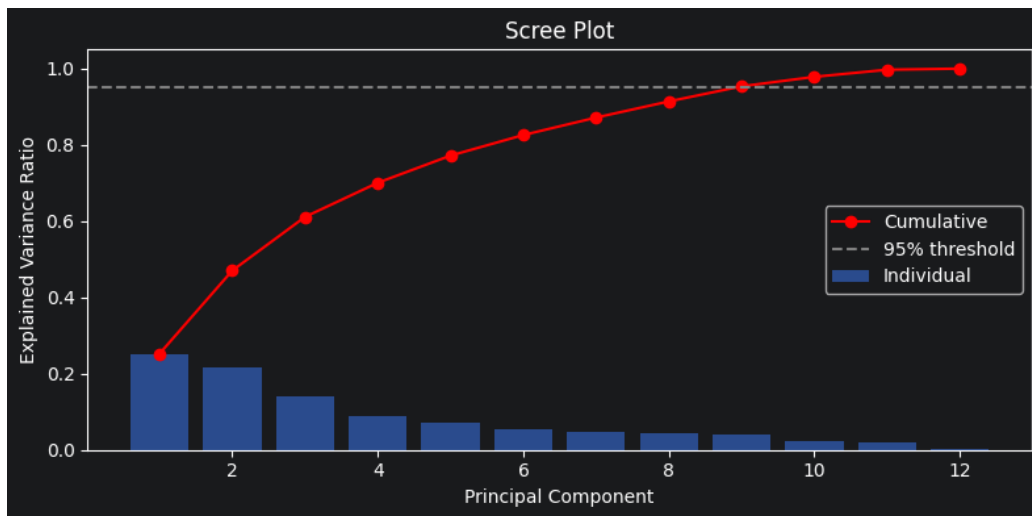


Figure 5: Scree plot showing explained variance for each component and cumulative explained variance. The 95% line helps choose a suitable number of components.

7. LDA From Scratch and Comparison With sklearn

LDA was implemented from scratch by calculating each class mean, computing the within-class scatter matrices, and calculating the projection vector W that separates the two classes. The data was then projected onto the LDA direction, and a threshold between the projected means of red and white wine was used for classification.

Metric / Value	Scratch LDA	sklearn LDA
W shape	(12,)	-
Train LDA shape	(4,256,)	approximately (4,256, 1)
Test LDA shape	(1,064,)	approximately (1,064, 1)
Red projected mean	1.000866	-
White projected mean	0.994718	-
Threshold	0.997792	-
Accuracy	0.9962 = 99.62%	0.9953 = 99.53%
Correct / Total	1,060 / 1,064	1,059 / 1,064
Errors	4	5

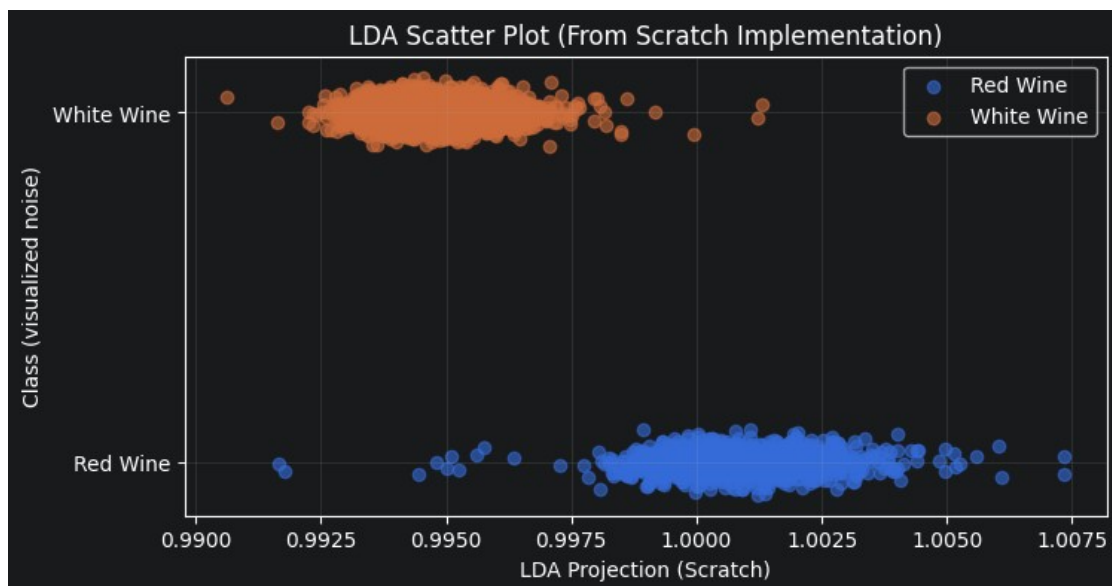


Figure 6: LDA projection from scratch. The plot shows a clear separation between red and white wine on one discriminant direction.

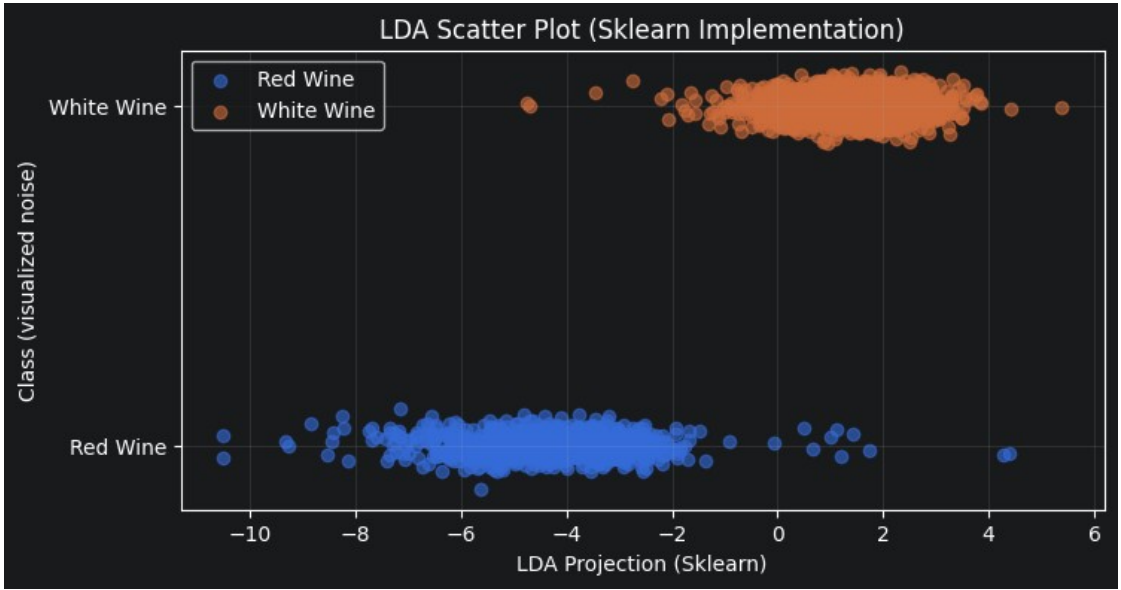


Figure 7: LDA projection using sklearn. This plot supports the same conclusion that the two wine types are strongly separable.

LDA Confusion Matrix

Model	True Red -> Pred Red	True Red -> Pred White	True White -> Pred Red	True White -> Pred White
Scratch LDA	269	3	1	791
sklearn LDA	268	4	1	791

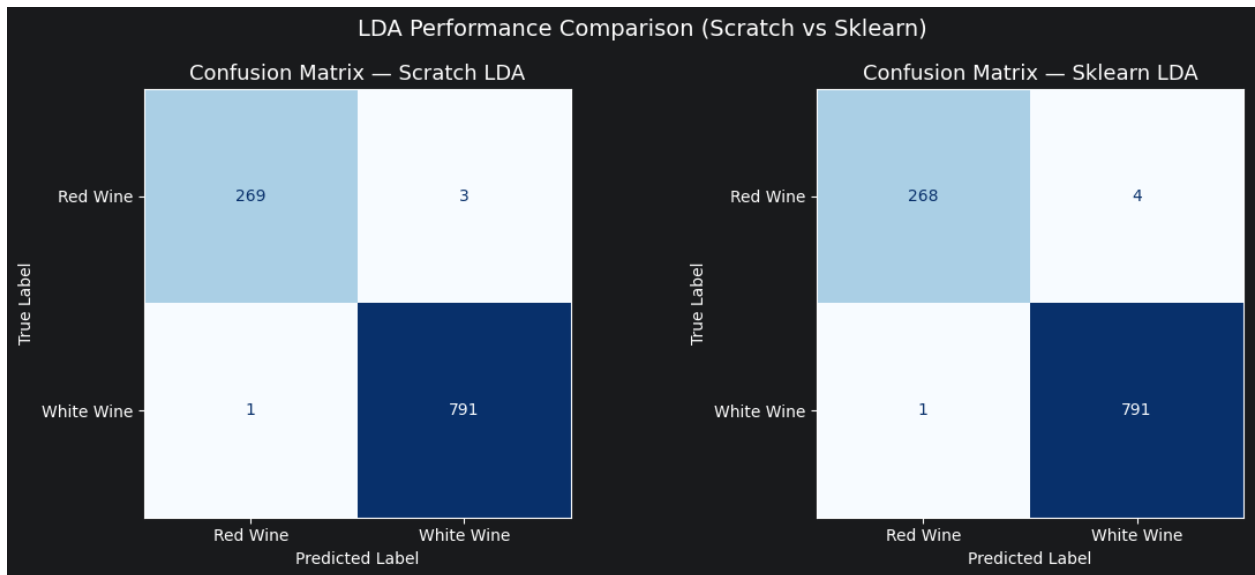


Figure 8: Confusion matrices for LDA. Scratch LDA made only 4 mistakes, while sklearn LDA made only 5 mistakes.

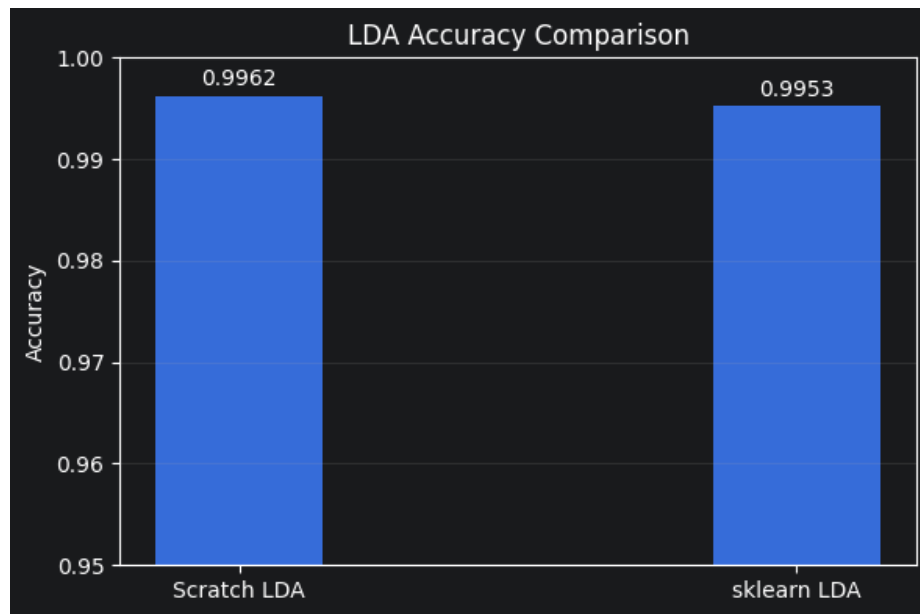


Figure 9: Accuracy comparison between Scratch LDA and sklearn LDA. Scratch LDA is slightly higher, but the difference is very small.

8. SVM With RBF Kernel

The RBF kernel was used because the previous visualizations showed that the classes are strongly separable, and there was no clear need for a polynomial relationship. The from-scratch SVM used labels as -1 and +1, calculated the RBF kernel between samples, and updated weights and bias during training. The result was then compared with sklearn SVC using the same RBF kernel idea.

Metric	SVM From Scratch	sklearn SVM
Kernel	RBF	RBF
Accuracy	0.9944 = 99.44%	0.9953 = 99.53%
Correct / Total	1,058 / 1,064	1,059 / 1,064
Errors	6	5

SVM Confusion Matrix

Model	True Red -> Pred Red	True Red -> Pred White	True White -> Pred Red	True White -> Pred White
Scratch SVM	267	5	1	791
sklearn SVM	268	4	1	791

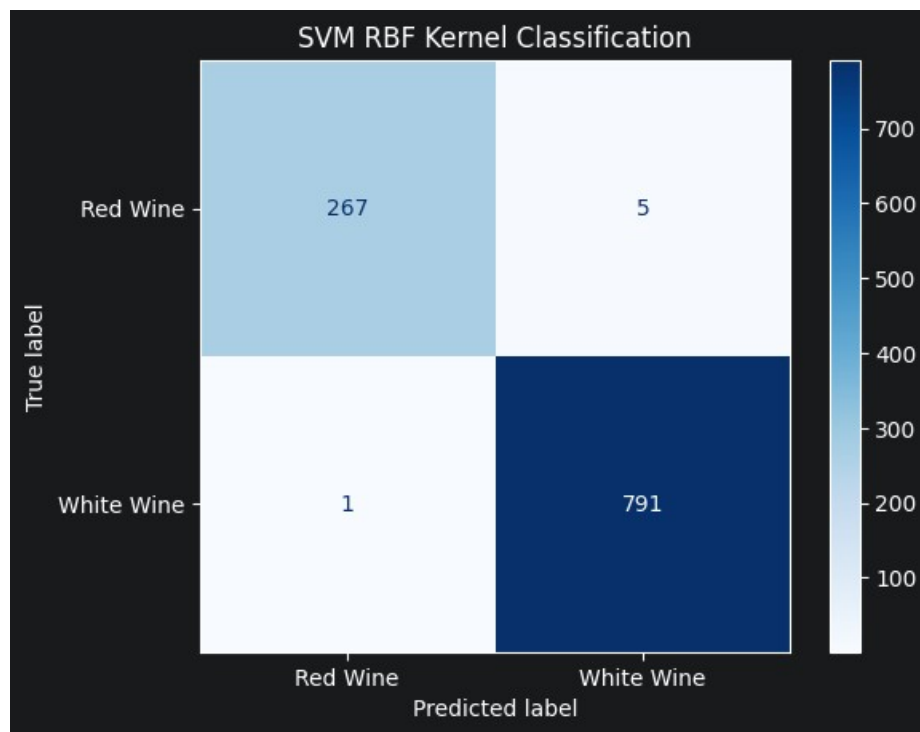


Figure 10: Confusion matrix for SVM from scratch. The model made 6 mistakes out of 1,064 test samples.

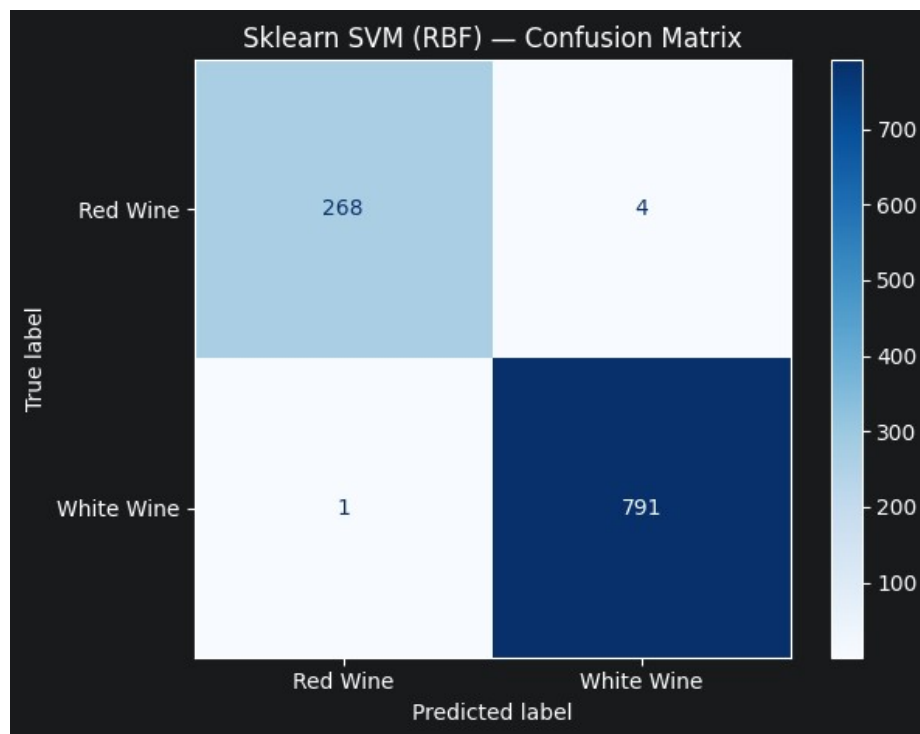


Figure 11: Confusion matrix for sklearn SVM. The model made only 5 mistakes.

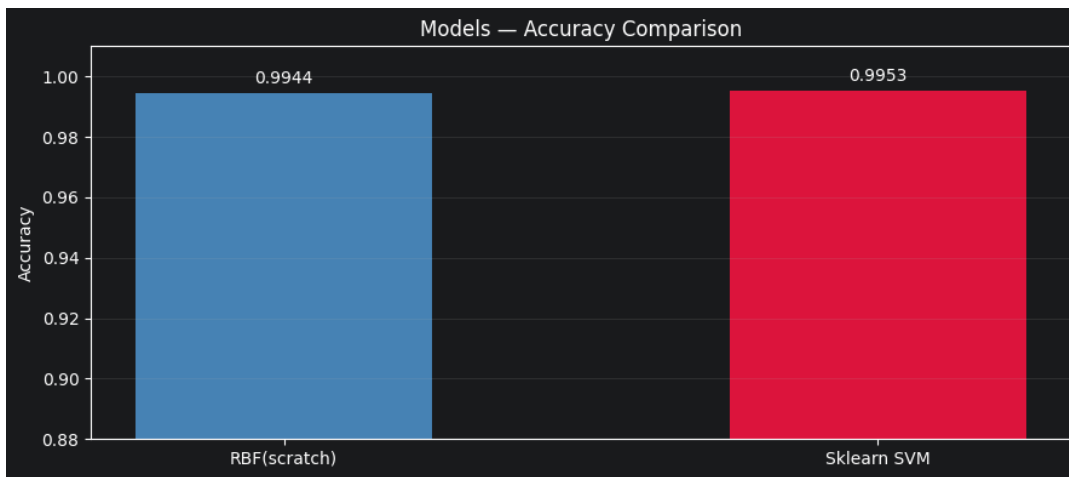


Figure 12: Accuracy comparison between SVM from scratch and sklearn SVM. sklearn is slightly higher, but the gap is very small.

9. Final Model Comparison

Model	Accuracy	Correct	Wrong	Note
Scratch LDA	99.62%	1,060	4	Best result in the project
sklearn LDA	99.53%	1,059	5	Very close to Scratch LDA
Scratch SVM RBF	99.44%	1,058	6	Slightly lower, but still excellent
sklearn SVM RBF	99.53%	1,059	5	Almost equal to sklearn LDA

All models achieved very high accuracy, which makes sense because the chemical differences between red wine and white wine are clear in the dataset. The best result was achieved by LDA from scratch with 99.62% accuracy. However, the differences between the models are very small, so it would not be fair to claim that one model is dramatically better than the others. Overall, all models performed extremely well on this split.

10. Conclusion

The project successfully covers the major stages of a machine learning workflow: reading the data, checking missing values and duplicates, performing exploratory data analysis with visualizations, scaling and handling outliers, implementing PCA from scratch and comparing it with sklearn, implementing LDA from scratch and comparing it with sklearn, and finally implementing SVM with an RBF kernel and comparing it with sklearn.

The final results show that the models can classify the wine type with accuracy above 99%. The strongest part of the project is the comparison between from-scratch implementations and sklearn implementations, because the similar results prove that the manual algorithm steps were implemented correctly. A possible future improvement would be to add cross-validation instead of relying on one train/test split and to include additional metrics such as precision, recall, and F1-score, especially because the dataset is imbalanced in favor of white wine.